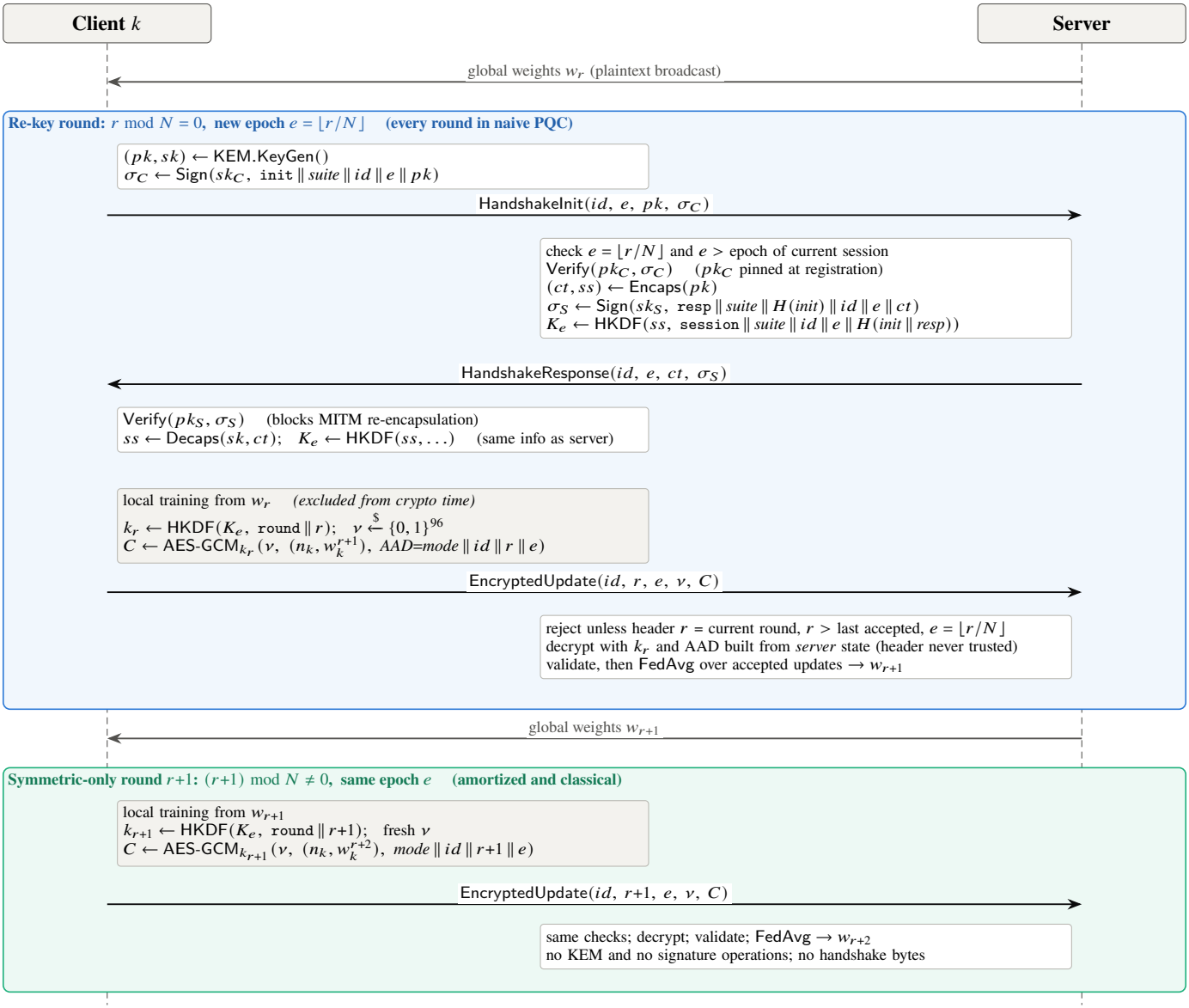
Client k
Server
global weights w_r (plaintext broadcast)
Re-key round: $r \bmod N = 0$, new epoch $e = \lfloor r/N \rfloor$ (every round in naive PQC)
$(pk, sk) \leftarrow \mathsf{KEM.KeyGen}()$
$\sigma_C \leftarrow \mathsf{Sign}(sk_C,\ \texttt{init} \,\|\, suite \,\|\, id \,\|\, e \,\|\, pk)$
$\mathsf{HandshakeInit}(id,\ e,\ pk,\ \sigma_C)$
check $e = \lfloor r/N \rfloor$ and $e >$ epoch of current session
$\mathsf{Verify}(pk_C, \sigma_C)$ (pk_C pinned at registration)
$(ct, ss) \leftarrow \mathsf{Encaps}(pk)$
$\sigma_S \leftarrow \mathsf{Sign}(sk_S,\ \texttt{resp} \,\|\, suite \,\|\, H(init) \,\|\, id \,\|\, e \,\|\, ct)$
$K_e \leftarrow \mathsf{HKDF}(ss,\ \texttt{session} \,\|\, suite \,\|\, id \,\|\, e \,\|\, H(init \,\|\, resp))$
$\mathsf{HandshakeResponse}(id,\ e,\ ct,\ \sigma_S)$
$\mathsf{Verify}(pk_S, \sigma_S)$ (blocks MITM re-encapsulation)
$ss \leftarrow \mathsf{Decaps}(sk, ct)$; $K_e \leftarrow \mathsf{HKDF}(ss, \ldots)$ (same info as server)
local training from w_r (excluded from crypto time)
$k_r \leftarrow \mathsf{HKDF}(K_e,\ \texttt{round} \,\|\, r)$; $\nu \xleftarrow{\$} \{0,1\}^{96}$
$C \leftarrow \mathsf{AES\text{-}GCM}_{k_r}(\nu,\ (n_k, w_k^{r+1}),\ AAD{=}mode \,\|\, id \,\|\, r \,\|\, e)$
$\mathsf{EncryptedUpdate}(id,\ r,\ e,\ \nu,\ C)$
reject unless header r = current round, $r >$ last accepted, $e = \lfloor r/N \rfloor$
decrypt with k_r and AAD built from server state (header never trusted)
validate, then FedAvg over accepted updates $\rightarrow w_{r+1}$
global weights w_{r+1}
Symmetric-only round $r{+}1$: $(r{+}1) \bmod N \neq 0$, same epoch e (amortized and classical)
local training from w_{r+1}
$k_{r+1} \leftarrow \mathsf{HKDF}(K_e,\ \texttt{round} \,\|\, r{+}1)$; fresh ν
$C \leftarrow \mathsf{AES\text{-}GCM}_{k_{r+1}}(\nu,\ (n_k, w_k^{r+2}),\ mode \,\|\, id \,\|\, r{+}1 \,\|\, e)$
$\mathsf{EncryptedUpdate}(id,\ r{+}1,\ e,\ \nu,\ C)$
same checks; decrypt; validate; FedAvg $\rightarrow w_{r+2}$
no KEM and no signature operations; no handshake bytes